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Ch. 7. Entropy and free energy

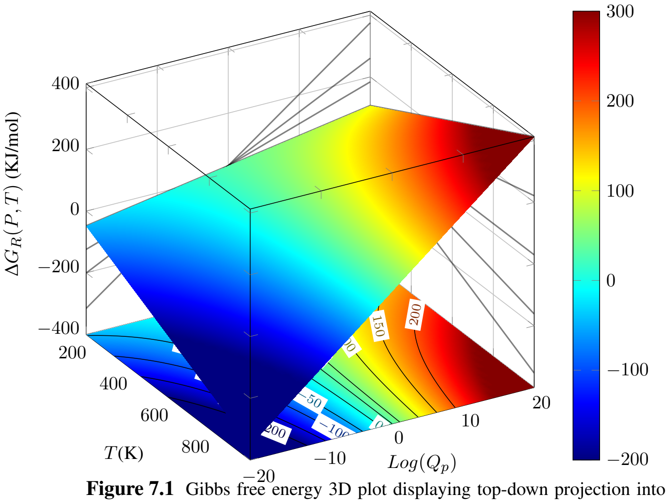

Figure 7.1 Gibbs free energy 3D plot displaying top-down projection into the LogQ_p-T bottom plane giving ΔG isolines, into the ΔG-T, right plane giving isobars with constant Q_p, and into the ΔG-LogQ_p, left plane giving isotherms with constant T. The color scale is blue for negative, small ΔG_R values, changing into red for positive, large ΔG values.

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7.1 Temperature and pressure effects on a reaction

Chemical reactions with positive values of ΔG_R are not spontaneous. Sometimes, we can adjust pressure or temperature to help spontaneity hence making ΔG_R negative. This section addresses the impact of external pressure and temperature conditions on spontaneity by using Gibbs free energy 3D and 2D contour maps. These graphs, similar to weather maps used to visualize atmospheric data, are useful to understand how pressure and temperature impact reactions of different thermal character.

`A two-dimensional function` $\Delta G_R(P,T)$ is a complex, two-dimensional function that changes with pressure and temperature. These two variables p and T are referred to as the external conditions of the reaction and can be controlled during an experiment. Differently, internal conditions, such as the enthalpy or entropy of reaction (ΔH° and ΔS°), only depend on the interaction between reactants and products. $\Delta G_R(P,T)$ data can be graphed in either as a 3D plots or as simplified-2D contour maps. On one hand, 3D $\Delta G_R(P,T)$ plots are graphical 3D representations of the relationship between $\Delta G_R(P,T)$, p, and T (see inclined, colored plot in Figure 7.1). On the other hand, 2D contour maps are two-dimensional visualizations representing $\Delta G_R(P,T)$ data using lines of equal value, often called isolines or contour lines (see

Figure 7.1 bottom, left and right projections). 2D contour maps are generated by taking 3D $\Delta G_R(P,T)$ plots and projecting their constant height lines onto a 2D plane. These 2D plots use contour lines to represent 3D topographical data, where parallel slices at different values represent constant output values. For example, Figure 7.1 displays a top-bottom into the T-LogQ_p plane, a left-to-right projection into the ΔG_R-T plane and a right-to-left projection into the ΔG_R-LogQ_p plane. 2D contour maps are widely used in scientific reporting, for example, in weather forecasting and meteorological analysis, to visualize complex weather patterns.

`Isolines` As we can project $\Delta G_R(P,T)$ in three different axis we can find three different types of isolines. First, if we project onto the LogQ_p-T plane, the bottom of the box in Figure 7.1, we obtain isolines with constant ΔG_R value, called *isoenergy lines*. These are curved lines with either a positive or a negative curve. A very important isoenergy line is the *equilibrium isoline*, that is, the isoenergy corresponding to a $\Delta G_R(P,T)$ value of zero. Second, if we project onto the ΔG-T plane, the right side of the box in Figure 7.1, we obtain isolines with constant LogQ_p value called *isobars* as Q_p depends on the pressure of reactants and products. These are straight lines with either a positive or a negative slope. Third, if we project onto the ΔG-LogQ_p plane, the left side of the box in Figure 7.1, we obtain isolines with constant T value called *isotherms*. These are straight lines, always with a positive slope.

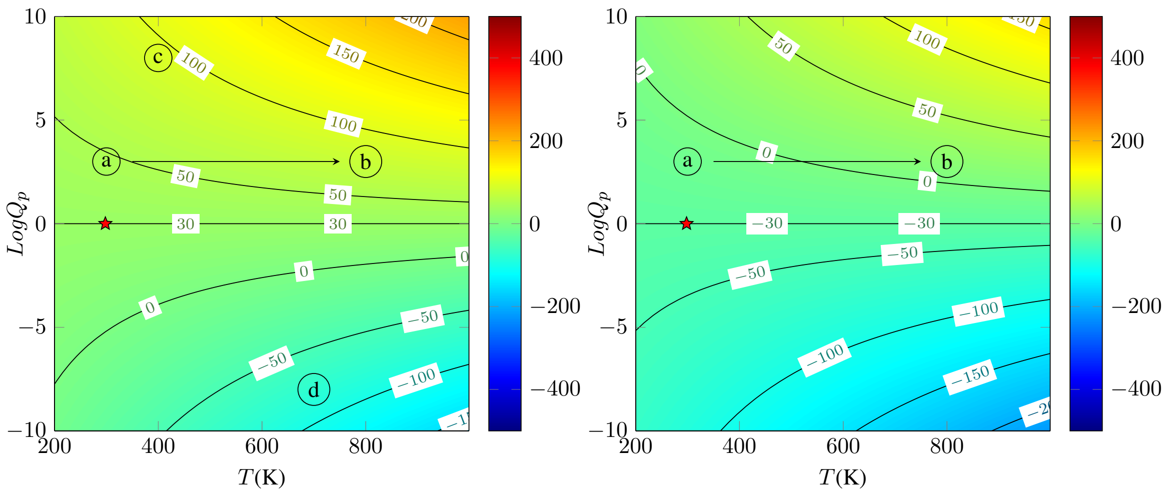

Figure 7.2 Gibbs free energy 3D plot displaying contour lines for different pressure and temperature conditions for an endergonic (Left Panel) and an exergonic (Right Panel) reaction. The star sign indicated the standard pressure and temperature conditions. The equilibrium isoline corresponds to $\Delta G_R = 0$. The reaction on the left is non-spontaneous under the conditions labeled as *a* on the left panel ($\Delta G_R > 0$), whereas the reaction on the right is spontaneous under the same conditions ($\Delta G_R < 0$). Increasing the temperature until reaching the conditions labeled as *b* does not affect the spontaneity of the reaction on the left, whereas it impedes the reaction on the right, which is now non-spontaneous under b conditions ($\Delta G_R > 0$).

`Interpretting contour maps` $\Delta G_R(P,T)$ contour maps, in particular those representing isoenergy lines, are useful graphical constructs. Information such as spontaneity, the thermal character (e.g., whether the reaction is endothermic or exothermic), or the entropic character (e.g., whether the reaction is producs of consumes entropy) of a reaction can be easily obtained just by inspecting the map. Let us first address spontaneity. We have that under the temperature and pressure conditions represented in point c in Figure 7.2 (left panel), we have that the reaction is nonspontaneous as $\Delta G_R(P,T)$ is positive. The reaction is hence *endergonic*. Differently, under the temperature and pressure conditions represented in point d the reaction will proceed spontaneously as $\Delta G_R(P,T)$ is now negative. The reaction is now *exergonic* The curvature of the equilibrium isoline informs about the thermal character of a reaction. Regarding the thermal character, we have that the reaction depicted in Figure 7.2 (left panel) is endothermic, as the equilibrium isoline bends upwards, whereas the reaction depicted in Figure 7.2 (right panel) is exothermic, as the equilibrium isoline bends downwards. The Asymptotic values of the equilibrium isoline for large temepratures informs about the entropic character of a reaction. We can see in Figure 7.2 (left panel) that the equilibrium isoline tends to a positive value at very large temperatures, hence the reaction will produce entropy. Differently, the reaction depicted in Figure 7.2 (right panel) consumes entropy as the equilibrium isoline tends to a negative value at very large temperatures.

`Pressure effects` Let us now address the impact of pressure on spontaneity. We can consider many different pressure conditions, that is, various combinations of reactant and product pressures. Let us first address two critical pressure conditions: the standard pressure conditions and the equilibrium pressure conditions.

On one hand, at *standard pressure conditions* all reactant and product pressures are 1atm and hence LogQ_c is zero. In these conditions $\Delta G_R(P,T)$ equals to $\Delta G^\circ_R(T)$. Standard conditions represent a single point on the Gibbs free energy two-dimensional plot (Figure 7.2, star mark). Under these conditions, $\Delta G_R(P,T)$ can be either positive or negative, depending on the tabulated standard Gibbs free energy values.

On the other hand, in *equilibrium pressure conditions*, all reactant and product pressures correspond to their corresponding equilibrium value and therefore Q_c, the pressure ratio, equals K_c, the real equilibrium constant. Under these pressure conditions, the reaction has reached equilibrium. In these conditions, we also have that ΔG_R will be zero. The equilibrium pressure conditions do not represent a point but an isoline in the $\Delta G^\circ_R(T)$-3D plot. Along this line, ΔG_R=0, giving a set of temperature and pressure conditions. This isoline is called the equilibrium isoline. Remember that K_c changes with temperature and hence there is not a single value for K_c for a reaction. We can, for example, obtain K_c for standard temperature conditions (T=298K, the temperature at which all thermodynamic data is tabulated) as shown in Figure 7.2. The equilibrium isoline either increases or decreases with temperature. It increases for endothermic reactions, and it decreases for exothermic reactions.

For all other pressure conditions, different than the standard and equilibrium pressure conditions, $\Delta G_R(P,T)$ can be either positive or negative, depending on the relative ratio between product and reactant pressures and on temperature. On one hand, for pressure conditions having more products (and fewer reactants) than in equilibrium, $\Delta G_R(P,T)$ will be positive: the reaction would not be spontaneous towards the right. This results from the fact that in these conditions $Q_p > 1$. On the other hand, for pressure conditions having fewer products (and more reactants) than in equilibrium, $\Delta G_R(P,T)$ will be negative: the reaction would be spontaneous towards the right. This results from the fact that in these conditions $Q_p < 1$. Still, we could always change the reactant and product pressure to help reach equilibrium. Now, let us address the effect of changing the pressure. Increasing the product's pressure, that is going from bottom to top of Figure 7.2, will make $\Delta G_R(P,T)$ more positive, whereas increasing the reactant's pressure, that is going from top to bottom of Figure 7.2, will make $\Delta G_R(P,T)$ more negative. This is a thermodynamic interpretation of the *Le Châtelier principle*: when we have many reactants and few products, the reaction will proceed to the right. Therefore, a very important idea to remember is that adding fewer products and more reactants than at equilibrium will *always* drive the reaction towards the right, helping spontaneity.

`Temperature effects` Let us address the dependence of Gibbs free energy on temperature by describing how ΔG_R isolines changes on Figure 7.2. For low temperatures, on the left side of each plot in Figure 7.2, isolines curve while ΔG_R increases. For high temperatures, on the right side of each plot in Figure 7.2, the impact of temperature is minimal and isolines flatten.

Unfortunately, the dependence of Gibbs free energy on temperature is tied to the LogQ_c value and we do not favour spontaneity just by increasing tempertaure. For example in Figure 7.2 (Left panel) we have that the reaction is nonspontaneous in both a and b temperature conditions. That is increasing temperature did not help spontaneity. Differently, on Figure 7.2 (Right panel) we have that the reaction is only spontaneous in b temperature conditions and not in a. That is increasing temperature did help spontaneity. Sellecting the right value for LogQ_c is important. For very small LogQ_c values, the isoenergy lines increase with temperature, curving upwards. This means that when the temperature increases, ΔG_R becomes more negative, hence favouring spontaneity. For large LogQ_c values, the isoenergy lines decrease with temperature, curving downwards. This means that when the temperature increases, ΔG_R becomes less negative, hence disfavouring spontaneity. There is a critical LogQ_c value that separates isolines curving upwards and downwards, corresponding to

$$\boxed{\text{Log}Q_c(critical) = \frac{\Delta S^\circ}{R}} \tag{7.1}$$

Only for LogQ_c values smaller than Log$Q_c(critical)$, we can favour spontaneity by increasing temperature. On Figure 7.2 (Left panel) Log$Q_c(critical)$ is approximately -3, whereas on the right panel, approximately +3. Still, even when LogQ_c is small than the critical value, temperature has to be large enough for a reaction to proceed spontaneusly. For low temperature, even when we do not cross Log$Q_c(critical)$ the reaction will not proceed. The critical temperature value is given by

$$\boxed{T(critical) = \frac{\Delta H^\circ}{\Delta S^\circ - R\text{Log}Q_c}} \tag{7.2}$$

The information here presented demonstrates how a combination of low enough pressure conditions and large enough temperatures need to be reached for a reaction to happen spontaneously. Mind that while the critical temperature value depends on ΔH°, ΔS° and LogQ_c, the critical LogQ_c value only depends on ΔS°.

`Interpretting 3D maps` Figure 7.3 displays a series of 3D Gibbs free energy maps. We can see that these are not flat maps, and the maps twist upwards in one extreme and downwards in the other. That is why the isobars, obtained by flattening the maps in the T-ΔG_R plane shown in the inserts, have different slopes of increasing value from negative to positive.

The gold here is to understand how these maps change with enthalpy and entropy. First, if we compare Figure 7.3 (Top, Left panel) with (Top, Right panel), we would see that decreasing enthalpy implies a downward shifting of the maps. A similar effect is found when comparing Figure 7.3 (Bottom, Left panel) with (Bottom, Right panel). That is the reason why exothermic reactions are often more likely to be spontaneous.

Differently, if we compare Figure 7.3 (Top, Left panel) with (Bottom, Left panel), we would see that increasing entropy implies a flattening of the isobars with positive slope, while pushing down the isobars with negative slope, hence making the plot more negative. A similar change is found when comparing Figure 7.3 (Top, Right panel) with (Bottom, Right panel). That is the reason why reactions with entropy production are often more likely to be spontaneous.

Summary In summary, the pressure and temperature conditions are tangled in their dependence on $\Delta G_R(P,T)$, whereas only temperature affects $\Delta G^\circ_R(T)$. We can always adjust the pressure ratio to decrease $\Delta G_R(P,T)$, helping spontaneity. Pressure effects always follow the same trend: adding more products and fewer reactants than at equilibrium will always help spontaneity. Differently, the impact of temperature on equilibrium varies, depending on the sign of $\Delta G^\circ_R(T)$: for negative $\Delta G^\circ_R(T)$ values increasing temperature helps spontaneity, whereas for positive $\Delta G^\circ_R(T)$ values decreasing temperature helps spontaneity.

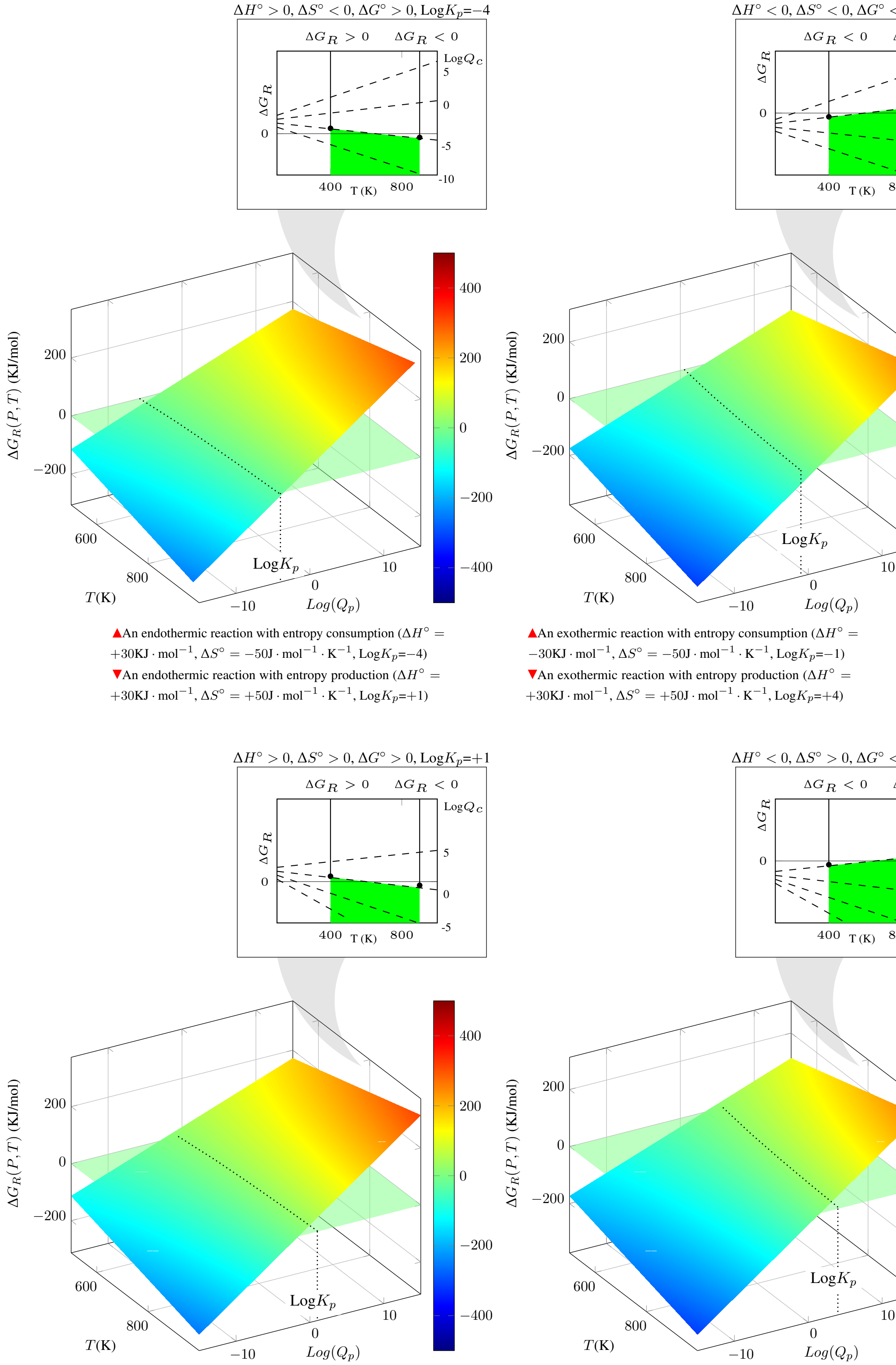

Figure 7.3 Gibbs free energy 3D plot for a set of reaction with different internal thermodynamics parameters: (Top Left) an endergonic reaction: you can increase T to favour the reaction only for pressure conditions (LogQ_c) lower than $\Delta S/R$); (Top Right) an exergonic reaction mediated by H: you disfavour the reaction by increasing T only for pressure conditions (LogQ_c) higher than $\Delta S/R$); (Bottom Left) an endergonic reaction mediated by S: you can increase T to favour the reaction only for pressure conditions (LogQ_c) lower than $\Delta S/R$); (Bottom Right) an exergonic reaction mediated by both the H and S: you disfavour the reaction by increasing T only for pressure conditions (LogQ_c) higher than $\Delta S/R$);

Table 7.1 Standard thermodynamic functions at 1atm and 298K.*

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
Al			
$Al_{(s)}$	0	0	28.3
$Al^{3+}_{(g)}$	5483.9	NA†	149.9
$Al_{(aq)}$	-524.7	-481.2	—
$AlF_{3(s)}$	-1504.1	-1425.1	66.4
$AlCl_{3(s)}$	-704.2	-628.9	110.7
$AlCl_3 \cdot 6\,H_2O_{(s)}$	-2691.6	NA†	NA†
$AlBr_{3(s)}$	-527.2	-488.4	163.2
$AlI_{3(s)}$	-313.8	-300.8	159.0
$Al_2O_{3(s)}$	-1675.7	-1582.4	50.9
$Al(OH)_{3(s)}$	-1287.4	-1149.8	85.4
$Al(NO_3)_3 \cdot 6\,H_2O_{(s)}$	-2850.5	-2203.9	467.8
$Al_2S_{3(s)}$	-723.8	NA†	decomp.
$Al_2(SO_4)_{3(s)}$	-3440.0	-3100.1	239.3
$Al_2(SO_4)_3 \cdot 6\,H_2O_{(s)}$	-5311.7	-4622.6	469.0
$Al_2(SO_4)_3 \cdot 18\,H_2O_{(s)}$	-8878.9	-7437.5	—
Sb			
$Sb^{3+}_{(g)}$	2703.3	NA†	168.7
$SbH_{3(g)}$	145.1	147.7	232.7
$SbF_{3(s)}$	-915.5	-807.0	105.4
$SbCl_{3(s)}$	-382.2	-323.7	184.0
$SbCl_{5(l)}$	-440.2	-350.2	301.0
$Sb_4O_{6(s)}$	-1440.6	-1268.2	220.9
$Sb_2S_3(black)_{(s)}$	-174.9	-173.6	182.0
$Sb_2(SO_4)_{3(s)}$	-2402.5	NA†	
As			
$As_{(s)}$	0	0	35.1
$As^{3+}_{(g)}$	5950.2	NA†	162.3
$AsH_{3(g)}$	66.4	68.9	222.7
$AsF_{3(l)}$	-956.3	-909.1	181.2
$AsF_{3(g)}$	-920.6	-905.7	289.0
$AsCl_{3(l)}$	-305.0	-259.4	216.3
$AsBr_{3(s)}$	-197.5	-169.0	161.1
$As_2O_{3(s)}$	-653.0	-571.0	117.0
$As_2O_{5(s)}$	-924.9	-782.4	105.4
$As_2O_{3(s)}$	-169.0	-168.6	163.6
$As_4O_{6(s)}$	-1314.0	-1153.0	223.0
Ba			
$Ba_{(s)}$	0	0	66.9
$Ba^{2+}_{(g)}$	1660.5	NA†	170.2
$Ba^{2+}_{(aq)}$	-537.0	-560.8	9.6
$BaH_{2(s)}$	-178.7	-132.2	NA†
$BaF_{2(s)}$	-1207.1	-1156.9	96.4
$BaCl_{2(s)}$	-858.6	-810.4	123.7
$BaCl_2 \cdot 2\,H_2O_{(s)}$	-1406.1	-1296.5	202.9
$Ba(ClO_3)_{2(s)}$	-762.7	-556.9	231.0
$Ba(ClO_3)_2 \cdot H_2O_{(s)}$	-1069.0	—NA†	0.125
$Ba(ClO_4)_{2(s)}$	-800.0	-535.1	249.0
$BaBr_{2(s)}$	-757.3	-736.8	146.0
$BaBr_2 \cdot 2\,HO_{(s)}$	-1366.1	-1230.5	226.0

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$Ba(BrO_3)_{2(s)}$	-752.7	-577.4	243.0
$Ba(BrO_3)_2 \cdot H_2O_{(s)}$	-1054.8	-824.6	292.5
$BaI_{2(s)}$	-602.1	-609.0	167.0
$BaI_2 \cdot 2\,H_2O_{(s)}$	-1216.7	NA†	0.63
$Ba(IO_3)_{2(s)}$	-1027.2	-864.8	249.0
$Ba(IO_3)_2 \cdot H_2O_{(s)}$	-1322.1	-1104.2	297.0
$BaO_{(s)}$	-553.5	-525.1	70.4
$BaO_{2(s)}$	-634.3	-572.0	65.7
$Ba(OH)_{2(s)}$	-944.7	-855.2	99.7
$BaCO_{3(s)}$	-1216.3	-1137.6	112.1
$Ba(HCO_3)_{2(s)}$	-1921.6	-1734.3	192.0
$Ba(NO_3)_{2(s)}$	-992.1	-796.7	213.8
$BaS_{(s)}$	-460.0	-456.0	78.2
$BaSO_{4(s)}$	-1473.2	-1362.3	151.9
$BaCrO_{4(s)}$	-1428.0	-1338.8	132.2
$BaC_2O_{4(s)}$	-1368.6	NA†	5.2×10^{-5}
$BaC_2O_4 \cdot 2\,H_2O_{(s)}$	-1971.1	NA†	5.20×10^{-5}
Be			
$Be_{(s)}$	0	0	9.5
$Be^{2+}_{(s)}$	2993.0	NA†	136.2
$BeF_{2(s)}$	-1026.8	-979.5	53.2
$BeCl_{2(s)}$	-490.4	-445.6	82.7
$BeCl_2 \cdot 4\,H_2O_{(s)}$	-1808.3	-1563.0	243.1
$BeBr_{2(s)}$	-353.5	-354.0	112.1
$BeO_{(s)}$	-609.6	-580.3	14.1
$Be(OH)_{2(s)}$	-902.4	-815.0	51.9
$Be(NO_3)_2 \cdot 3\,H_2O_{(s)}$	-787.8	NA†	0.804
$BeS_{(s)}$	-234.3	-232.0	35.0
$BeSO_{4(s)}$	-1205.2	-1093.9	77.9
$BeSO_4 \cdot 4\,H_2O_{(s)}$	-2423.7	-2080.7	234.0
Bi			
$Bi_{(s)}$	0	0	56.9
$Bi^{3+}_{(g)}$	5005.7	NA†	
$BiCl_{3(s)}$	-379.1	-315.1	177.0
$Bi(ClO)_{3(s)}$	-366.9	-322.2	120.5
$BiI_{3(s)}$	-105.0	-175.3	233.9
$Bi_2O_{3(s)}$	-573.9	-493.7	151.5
$Bi_2S_{3(s)}$	-143.1	-140.6	200.46
$Bi_2(SO_4)_{3(s)}$	-2544.3	-2583.6	NA†
B			
$B_{(s)}$	0	0	5.9
$B^{3+}_{(s)}$	7468.0	NA†	138.5
$B_2H_{6(g)}$	35.6	86.6	232.0
$BF_{3(g)}$	-137.0	-1120.3	254.0
$BCl_{3(l)}$	-427.2	-387.4	206.3
$BCl_{3(g)}$	-403.7	-388.7	290.0
$BI_{3(g)}$	71.1	20.8	349.1
$B_2O_{3(s)}$	-1272.8	-1193.7	54.0
$B_2O_{3(l)}$	-1254.5	-1182.4	77.8
$B(OH)_{3(s)}$	-1094.0	-969.0	88.8

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$BN_{(s)}$	-254.4	-228.4	14.8
$B_2S_{3(s)}$	-240.6	-229.0	57.4
Br			
$Br_{2(l)}$	0	0	152.2
$Br_{2(g)}$	30.9	3.1	245.4
$Br^{-1}_{(g)}$	-233.9	-238.7	163.4
Cd			
$Cd_{(s)}$	0	0	51.8
$Cd^{2+}_{(g)}$	2623.5	NA†-	167.7
$CdF_{2(s)}$	-700.4	-647.7	77.4
$CdCl_{2(s)}$	-391.5	-344.0	115.3
$CdCl_2 \cdot H_2O_{(s)}$	-688.4	-587.1	167.8
$Cd(ClO_4)_{2(aq)}$	-334.6	-94.8	290.8
$Cd(ClO_4)_2 \cdot 6\,H_2O_{(s)}$	-2052.7	NA†	
$CdBr_{2(s)}$	-316.2	-296.3	137.2
$CdI_{2(s)}$	-203.3	-201.4	161.1
$Cd(IO_3)_{2(s)}$	NA†-	377.1	NA†–
$CdO_{(s)}$	-258.2	-228.4	54.8
$Cd(OH)_{2(s)}$	-560.7	-473.6	96.0
$Cd(CN)_{2(s)}$	162.2	207.9	104.2
$Cd(NO_3)_{2(s)}$	-456.3	-259.0	197.9
$Cd(NO_3)_2 \cdot 2\,H_2O_{(s)}$	-1055.6	-748.9	NA†
$Cd(NO_3)_2 \cdot 4\,H_2O_{(s)}$	-1649.0	-1217.1	NA†
$CdS_{(s)}$	-161.9	-156.5	64.8
$CdSO_{4(s)}$	-933.3	-822.8	123.0
$CdSO_4 \cdot 2.67\,H_2O_{(s)}$	-1729.4	-1465.3	229.6
Cs			
$Cs_{(s)}$	0	0	
$Cs^{1+}_{(g)}$	458.0	NA†	169.7
$CsF_{(s)}$	-553.5	-525.5	92.8
$CsCl_{(s)}$	-443.0	-414.5	101.2
$CsClO_{3(s)}$	-411.7	-307.9	156.1
$CsClO_{4(s)}$	-443.1	-314.3	175.1
$CsBr_{(s)}$	-405.8	-391.4	113.1
$CsI_{(s)}$	-346.0	-340.6	123.1
$CsIO_{4(s)}$	NA†	-380.7	184.0
$Cs_2O_{(s)}$	-345.8	-308.2	146.9
$CsOH_{(s)}$	-417.2	-359.0	86.0
$CsHCO_{3(s)}$	-966.1	-831.8	130.0
$CsNO_{3(s)}$	-506.0	-406.6	155.2
$Cs_2SO_{4(s)}$	-1443.0	-1323.7	211.9
Ca			
$Ca_{(s)}$	0	0	41.4
$Ca^{2+}_{(g)}$	1925.0	NA†	154.8
$CaH_{2(s)}$	-186.2	-147.3	42.0
$CaF_{2(s)}$	-1219.6	-1167.3	68.9
$CaCl_{2(s)}$	-795.8	-748.1	104.6
$CaCl_2 \cdot H_2O_{(s)}$	-1109.2	-1010.9	NA†
$CaCl_2 \cdot 2\,H_2O_{(s)}$	-1402.9	NA†	0.665
$CaCl_2 \cdot 4\,H_2O_{(s)}$	-2009.6	-1724.0	212.6

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$CaCl_2 \cdot 6\,H_2O_{(s)}$	-2607.9	-2205.0	284.9
$Ca(ClO_4)_{2(s)}$	-736.8	NA†	233.0
$Ca(ClO_4)_2 \cdot 4\,H_2O_{(s)}$	-1948.9	-1476.8	433.5
$CaBr_{2(s)}$	-682.8	-663.6	130.0
$CaBr_2 \cdot 6\,H_2O_{(s)}$	-2506.2	-2153.1	410.0
$Ca(BrO_3)_{2(s)}$	-718.8	NA†	227.6
$CaI_{2(s)}$	-533.5	-528.9	142.0
$CaI_2 \cdot 8\,H_2O_{(s)}$	-2929.6	NA†	
$Ca(IO_3)_{2(s)}$	-1002.5	-893.3	230.1
$Ca(IO_3)_2 \cdot H_2O_{(s)}$	-1293.3	NA†	
$Ca(IO_3)_2 \cdot 6\,H_2O_{(s)}$	-2780.7	-2267.7	451.9
$CaO_{(s)}$	-635.1	-604.0	39.7
$Ca(OH)_{2(s)}$	-986.1	-898.6	83.4
$CaC_{2(s)}$	-59.1	-64.8	69.9
$CaCO_{3(s),calcite}$	-1206.9	-1128.8	92.9
$CaCO_{3(s),aragonite}$	-1207.1	-1127.8	88.7
$Ca(NO_3)_{2(s)}$	-635.1	-743.2	193.3
$Ca(NO_3)_2 \cdot 2\,H_2O_{(s)}$	-1540.8	-1229.3	269.4
$Ca(NO_3)_2 \cdot 3\,H_2O_{(s)}$	-1838.0	-1471.9	319.2
$Ca(NO_3)_2 \cdot 4\,H_2O_{(s)}$	-2132.3	-1713.5	375.3
$CaS_{(s)}$	-482.4	-477.4	56.5
$CaSO_{3(s)}$	-1156.0	NA†	
$CaSO_{4(s)}$	-1431.1	-1321.9	106.7
$CaSO_4 \cdot 0.5\,H_2O_{(s)}$	-1576.7	-1436.8	130.5
$CaSO_4 \cdot 2\,H_2O_{(s)}$	-2022.6	-1797.4	194.1
$Ca_3(PO_4)_{2(s)}$	-4120.8	-3884.8	236.0
$CaCrO_4 \cdot 2\,H_2O$	-1379.0	-1277.4	133.9
$CaC_2O_{4(s)}$	-1360.6	NA†	NA†
$CaC_2O_4 \cdot H_2O_{(s)}$	-1674.9	-1514.0	156.5
$CaSi_{2(s)}$	-151.0	NA†	decomp.
$CaSiO_{3(s)}$	-1634.9	-1549.7	81.9
$Ca_2SiO_{4(s)}$	-2307.5	-2192.8	127.7
C(s), graphite	0	0	5.7
C(s), diamond	1.9	2.9	2.4
$C_{(g)}$	716.7	671.3	158.0
$C_{2(g)}$	836.8	780.4	199.3
$C_{3(g)}$	793.5	773.1	212.1
$CCl_{4(l)}$	-134.0	-65.3	214.4
$CO_{(g)}$	-110.5	-137.2	197.6
$CO_{2(g)}$	-393.5	-394.4	213.6
$CO_{2(aq)}$	-413.8	-386.0	117.6
$CO_3{}^{2-}{}_{(aq)}$	-677.1	-527.8	-56.9
$C_2N_{2(g)}$	307.9	296.3	242.1
$CS_{2(l)}$	98.7	65.2	151.3
$CS_{2(g)}$	117.0	67.2	237.7
Cl			
$Cl_{2(g)}$	0	0	233.0
$Cl^{-1}_{(g)}$	-246.0	-240.0	153.1
$Cl_2O_{(g)}$	80.3	97.9	266.1
$ClO_{2(g)}$	102.5	120.5	256.7

* Adapted from: https://issr.edu.kh/science/Reference_Tables_and_Values/10545-Nuffield%20Advance%20Science%20Book%20of%20Data.pdf

Table 7.1 (continued) Standard thermodynamic functions at 1atm and 298K.

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
Cr			
$Cr_{(s)}$	0	0	23.8
$Cr^{3+}_{(aq)}$	-232.0	–NA†	
$CrF_{3(s)}$	-1159.0	-1088.0	93.9
$CrCl_{2(s)}$	-326.0	-282.0	115.0
$CrCl_{3(s)}$	-556.5	-486.2	115.3
$CrO_2Cl_{2(l)}$	-579.5	-510.9	221.8
$CrI_{3(s)}$	-205.0	-202.5	NA†
$Cr_2O_{3(s)}$	-1139.7	-1058.1	81.2
$CrO_{3(s)}$	-598.5	-501.0	NA†
$Cr_2(SO_4)_{3(s)}$	-3025.0	NA†	0.163
$Cr_2(SO_4)_3 \cdot 18\,H_2O_{(s)}$	-8339.0	NA†	0.167
$Cr(CO)_{6(s)}$	-1076.9	-975.0	NA†
Co			
$Co_{(s)}$	0	0	30.0
$Co^{2+}_{(g)}$	2841.6	NA†	178.8
$CoF_{3(g)}$	-810.9	-707.0	94.6
$CoCl_{2(s)}$	-312.5	-269.9	109.2
$CoCl_2 \cdot 2\,H_2O_{(s)}$	-923.0	-764.8	188.0
$CoCl_2 \cdot 6\,H_2O_{(s)}$	-2115.4	-1725.5	343.0
$Co(ClO_4)_{2(aq)}$	-316.7	-71.5	251.0
$Co(ClO_4)_2 \cdot 6\,H_2O_{(s)}$	-2038.4	NA†	0.707
$CoBr_{2(s)}$	-220.0	-210.0	135.6
$CoBr_2 \cdot 6\,H_2O_{(s)}$	-2020.0	NA†	
$CoI_{2(s)}$	-88.7	-101.3	158.2
$Co(IO_3)_{2(aq)}$	-500.8	-310.4	125.5
$Co(IO_3)_2 \cdot 2\,H_2O_{(s)}$	-1081.9	-795.8	267.8
$CoO_{(s)}$	-237.9	-214.2	53.0
$Co_3O_{4(s)}$	-891.0	-774.0	102.5
$Co(OH)_{2(s)}$	-539.7	-454.4	79.0
$Co(NO_3)_{2(s)}$	-420.5	-237.0	192.0
$Co(NO_3)_2 \cdot 2\,H_2O_{(s)}$	-1021.7	NA†	
$Co(NO_3)_2 \cdot 3\,H_2O_{(s)}$	-1325.9	NA†	
$Co(NO_3)_2 \cdot 4\,H_2O_{(s)}$	-1630.5	NA†	
$Co(NO_3)_2 \cdot 6\,H_2O_{(s)}$	-2211.2	-1655.6	NA†
$CoS_{(s)}$	-80.8	-82.8	67.4
$CoSO_{4(s)}$	-888.3	-782.4	118.0
$CoSO_4 \cdot 7\,H_2O_{(s)}$	-2979.9	-2473.8	406.1
Cu			
$Cu_{(s)}$	0	0	33.2
$Cu^{2+}_{(g)}$	3054.0	NA†	179.0
$CuF_{2(s)}$	-542.7	-481.0	88.0
$CuF_2 \cdot 2\,H_2O_{(s)}$	NA†	-981.6	NA†
$CuCl_{(s)}$	-137.2	-119.9	86.2
$CuCl_{2(s)}$	-220.1	-175.7	108.1
$Cu(ClO_4)_{2(aq)}$	-193.1	48.3	264.4
$Cu(ClO_4)_2 \cdot 6\,H_2O_{(s)}$	-1928.4	NA†	very
$CuBr_{2(s)}$	-141.8	-108.7	118.0
$CuBr_2 \cdot 4\,H_2O_{(s)}$	-1326.3	-1081.1	293.7
$CuI_{(s)}$	-67.7	-69.5	96.7

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$Cu(IO_3)_{2(aq)}$	-377.8	-190.4	137.2
$Cu(IO_3)_2 \cdot H_2O_{(s)}$	-692.0	-468.6	247.2
$Cu_2O_{(s)}$	-168.6	-146.0	93.1
$CuO_{(s)}$	-157.3	-129.7	42.6
$Cu(OH)_{2(s)}$	-449.8	-359.4	75.0
$Cu(NO_3)_{2(s)}$	-302.9	-118.2	193.0
$Cu(NO_3)_2 \cdot 3\,H_2O_{(s)}$	-1217.1	NA†	0.570
$Cu(NO_3)_2 \cdot 6\,H_2O_{(s)}$	-2110.8	NA†	0.824
$Cu_2S_{(s)}$	-79.0	-86.2	120.9
$CuS_{(s)}$	-53.1	-53.6	66.5
$CuSO_{4(s)}$	-771.4	-661.9	109.0
$CuSO_4 \cdot 5\,H_2O_{(s)}$	-2279.6	-1880.1	300.4
F			
$F_{2(g)}$	0	0	202.7
$F^{-1}_{(g)}$	-270.7	-266.6	145.4
$F_2O_{(g)}$	-21.7	-4.7	247.3
Ga			
$Ga^{3+}_{(g)}$	5816.0	NA†	161.6
$GaF_{3(s)}$	-1163.0	-1085.3	84.0
$GaCl_{3(s)}$	-524.7	-454.8	142.0
$GaBr_{3(s)}$	-386.6	-359.8	180.0
$GaI_{3(s)}$	-238.9	-217.6	49.0
$Ga_2O_{3(s)}$	-1089.1	-998.3	85.0
Ge			
$Ge^{4+}_{(g)}$	10412.3	NA†	
$GeF_{4(g)}$	-NA†	302.8	decomp.
$GeCl_{2(s)}$	—NA†	NA†-	
$GeCl_{4(l)}$	-531.8	-462.8	245.6
$GeBr_{4(l)}$	-347.7	-331.4	280.7
$GeBr_{4(g)}$	-300.0	-318.0	396.1
$GeO_{(s)}$	-212.1	-237.2	50.0
$GeO_{2(s)}$	-551.0	-497.1	55.3
$GeS_{(s)}$	-69.0	-71.5	71.0
$GeS_{2(s)}$	-189.5	—NA†-	0.00329
Au			
$Au_{(s)}$	0	0	47.7
$Au^{1+}_{(g)}$	1262.4	NA†–	174.7
$AuH_{(g)}$	294.9	265.7	211.0
$AuF_{3(s)}$	-363.0	-297.5	210.9
$AuCl_{3(s)}$	-117.6	-55.2	147.3
$AuCl_3 \cdot 2\,H_2O_{(s)}$	-715.0	-519.0	226.0
$AuBr_{3(s)}$	-53.3	-31.0	100.0
$AuI_{(s)}$	0.0	-0.2	119.2
$Au_2O_{3(s)}$	-3.3	76.2	NA†–
H			
$H_{2(g)}$	0	0	130.6
$HF_{(g)}$	-271.1	-273.2	173.7
$HCl_{(g)}$	-92.3	-95.2	186.8
$HCl_{(aq)}$	-167.2	-131.2	56.5
$HClO_{(aq)}$	-131.3	-80.2	106.8

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$HBr_{(g)}$	-36.4	-53.4	198.6
$HI_{(g)}$	26.5	1.7	206.5
$HIO_{3(s)}$	-230.1	-144.3	118.0
$H_2O_{(l)}$	-285.8	-237.2	69.9
$H_2O_{(g)}$	-241.8	-228.6	188.7
$H_2O_{2(l)}$	-187.8	-120.4	109.6
$H_3AsO_{3(aq)}$	-742.2	-NA†	
$H_3AsO_{4(aq)}$	-902.5	NA†	
$HCN_{(l)}$	108.9	124.9	112.8
$HCN_{(g)}$	135.1	124.7	201.7
$H_2CO_{3(aq)}$	-699.6	-623.3	187.4
$HCO_3{}^{-1}{}_{(aq)}$	-692.0	-586.8	91.2
$HNO_{3(l)}$	-174.1	-80.8	266.3
$H_2S_{(g)}$	-20.6	-33.6	205.7
$H_2S_{(aq)}$	-39.7	-27.9	121.3
$H_2S_{2(l)}$	-23.1	NA†	decomp.
$H_2Se_{(g)}$	76.0	62.3	219.0
$H_2SO_{4(l)}$	-814.0	-690.1	156.9
$H_2SO_{4(aq)}$	-909.3	-744.5	20.1
$H_2Te_{(g)}$	154.0	138.0	234.0
$H_3PO_{4(s)}$	-1279.0	-1119.2	110.5
$H_3BO_{3(s)}$	-1094.3	-969.0	88.8
$H_3O_1{}^{+}{}_{(g)}$	979.9	NA†	
$OH^{+}_{(g)}$	1328.4	NA†	
$OH^{-}_{(g)}$	-140.9	NA†	
$H_2S^{+}_{(aq)}$	995.0	NA†	
I			
$I_{2(s)}$	0	0	116.1
$I_{2(g)}$	62.4	19.4	260.6
$IF_{(g)}$	-95.6	-118.5	236.1
$I_2{}^{+}{}_{(g)}$	967.5	NA†	
$ICl_{(s)}$	-35.1	NA†	decomp.
$ICl_{3(s)}$	-89.5	-22.3	167.4
$IBr_{(s)}$	-10.5	NA†	138.1
$I_2O_{5(s)}$	-158.1	-38.0	NA†
$I^{-1}_{(g)}$	-196.6	-221.9	169.1
Fe			
$Fe_{(s)}$	0	0	27.0
$Fe^{2+}_{(g)}$	2752.2	NA†	177.2
$Fe^{2+}_{(aq)}$	-89.1	-78.9	137.7
$Fe^{3+}_{(g)}$	-48.5	-4.7	315.9
$FeF_{2(s)}$	-686.0	-644.0	87.0
$FeF_{3(aq)}$	-1046.4	-841.0	357.0
$FeCl_{2(s)}$	-341.8	-302.3	117.9
$FeCl_2 \cdot 2\,H_2O_{(s)}$	-953.1	-797.5	NA†
$FeCl_2 \cdot 4\,H_2O_{(s)}$	-1549.3	-1275.7	NA†
$FeCl_{3(s)}$	-399.5	-334.1	142.3
$FeCl_3 \cdot 6\,H_2O_{(s)}$	-2223.8	-1812.9	NA†
$Fe(ClO_4)_{2(aq)}$	-347.7	-96.1	226.4
$Fe(ClO_4)_2 \cdot 6\,H_2O_{(s)}$	-2086.6	NA†	0.270

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$FeBr_{2(s)}$	-249.8	-236.0	140.7
$FeI_{2(s)}$	-113.0	-128.4	77.0
$FeI_{3(g)}$	71.0	NA†	
$FeO_{(s)}$	-271.9	-245.4	58.5
$Fe_2O_{3(s)}$	-824.2	-742.2	87.4
$Fe_3O_{4(s)}$	-1118.4	-1015.5	146.4
$Fe(OH)_{2(s)}$	-569.0	-486.6	88.0
$Fe(OH)_{3(s)}$	-823.0	-696.6	106.7
$FeCO_{3(s)}$	-740.6	-666.7	92.9
$Fe(CO)_{5(l)}$	-774.0	-705.4	338.1
$FeS_{(s)}$	-100.0	-100.4	60.3
$FeS_{2(s)}$	-178.2	-166.9	52.9
$FeSO_{4(s)}$	-928.4	-820.9	107.5
$FeSO_4 \cdot 7\,H_2O_{(s)}$	-3014.6	-2510.3	409.2
$Fe_{2(s)4)3(s)}$	-2581.5	NA†	261.7
$Fe(NO_3)_{3(aq)}$	-674.9	NA†	
$Pb_{(s)}$	0	0	64.8
$Pb^{2+}_{(g)}$	916.8	NA†	175.3
$Pb^{2+}_{(aq)}$	-1.7	-24.4	10.5
$PbF_{2(s)}$	-664.0	-617.1	110.5
$PbCl_{2(s)}$	-359.4	-314.1	136.0
$PbCl_{4(l)}$	-329.2	-259.0	NA†
$PbBr_{2(s)}$	-278.7	-261.9	161.5
$Pb(BrO_3)_{2(s)}$	-134.0	-50.0	NA†
$PbI_{2(s)}$	-175.5	-173.6	174.5
$PbO_{(s)}$	-217.3	-187.9	68.7
$PbO_{2(s)}$	-277.4	-217.4	68.6
$Pb(OH)_{2(s)}$	-515.9	-420.9	88.0
$Pb_3O_{4(s)}$	-718.4	-601.2	211.3
$PbCO_{3(s)}$	-700.0	-626.3	131.0
$Pb(NO_3)_{2(s)}$	-451.9	-251.0	213.0
$PbS_{(s)}$	-100.4	-98.7	91.2
$PbSO_{4(s)}$	-919.0	-813.2	148.6
$PbCrO_{4(s)}$	-899.6	-819.6	152.7
$Pb(CH_3COO)_2 \cdot 3\,H_2O_{(s)}$	-1851.0	-NA†	0.204
$Pb(C_2H_5)_{4(l)}$	52.7	336.4	472.5
Li			
$Li_{(s)}$	0	0	28.4
$Li^{+}_{(g)}$	679.6	650.0	132.9
$Li^{+}_{(aq)}$	-278.6	NA†	10.3
$LiH_{(s)}$	-90.5	-68.4	20.3
$Li_3H_{4(s)}$	NA†	NA†	decomp.
$LiF_{(s)}$	-616.0	-587.7	35.6
$LiCl_{(s)}$	-408.6	-384.4	59.3
$LiClO_{3(s)}$	-369.0	NA†	5.531
$LiClO_{4(s)}$	-381.0	NA†	0.564
$LiClO_4 \cdot H_2O_{(s)}$	-697.1	-509.6	155.2
$LiClO_4 \cdot 3\,H_2O_{(s)}$	-1298.0	-1001.3	254.8
$LiBr_{(s)}$	-351.2	-342.0	74.3
$LiBr \cdot H_2O_{(s)}$	-662.6	-594.3	109.6

Table 7.1 (continued) Standard thermodynamic functions at 1atm and 298K.

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$LiBr \cdot 2H_2O_{(s)}$	-962.7	-840.6	162.3
$LiBrO_{3(s)}$	-347.0	NA†	
$LiI_{(s)}$	-270.4	-270.3	86.8
$LiI \cdot H_2O_{(s)}$	-590.3	-531.4	123.0
$LiI \cdot 2H_2O_{(s)}$	-890.4	-780.3	184.0
$LiI \cdot 3H_2O_{(s)}$	-1192.1	NA†	0.804
$LiIO_{3(s)}$	-503.4	NA†	0.442
$Li_2O_{(s)}$	-597.9	-561.2	37.6
$LiOH_{(s)}$	-484.4	-439.0	42.8
$LiOH \cdot H_2O_{(s)}$	-788.0	-681.0	71.2
$Li_2CO_{3(s)}$	-1215.9	-1132.1	90.4
$LiHCO_{3(s)}$	-969.6	-880.9	123.4
$Li_3N_{(s)}$	-199.0	-155.4	37.7
$LiNO_{3(s)}$	-483.1	-381.2	90.0
$LiNO_3 \cdot 3H_2O_{(s)}$	-1374.4	-1103.7	223.4
$Li_2SO_{4(s)}$	-1436.5	-1321.8	115.1
$Li_2SO_4 \cdot H_2O_{(s)}$	-1735.5	-1565.7	163.6
$Li_3PO_{4(s)}$	-2095.8	NA†	0.000257
$LiAlH_{4(s)}$	-116.3	-44.8	78.7
	Mg		
$Mg_{(s)}$	0	0	32.5
$Mg^{2+}_{(aq)}$	-466.9	-454.8	-138.1
$MgF_{2(s)}$	-1123.4	-1070.3	57.2
$MgCl_{2(s)}$	-641.3	-591.8	89.6
$MgCl_2 \cdot H_2O_{(s)}$	-966.6	-861.8	137.2
$MgCl_2 \cdot 2H_2O_{(s)}$	-1279.7	-1118.1	179.9
$MgCl_2 \cdot 4H_2O_{(s)}$	-1898.9	-1623.5	264.0
$MgCl_2 \cdot 6H_2O_{(s)}$	-2499.0	-2115.0	366.1
$Mg(ClO_4)_{2(s)}$	-568.9	-432.2	213.0
$Mg(ClO_4)_2 \cdot 2H_2O_{(s)}$	-1218.7	NA†	
$Mg(ClO_4)_2 \cdot 4H_2O_{(s)}$	-1837.2	NA†	
$Mg(ClO_4)_2 \cdot 6H_2O_{(s)}$	-2445.5	-1863.1	520.9
$MgBr_{2(s)}$	-524.3	-503.8	117.2
$MgBr_2 \cdot 6H_2O_{(s)}$	-2410.0	-2056.0	397.0
$MgI_{2(s)}$	-364.0	-358.2	129.7
$MgO_{(s)}$	-601.7	-569.4	26.9
$Mg(OH)_{2(s)}$	-924.5	-833.6	63.2
$MgCO_{3(s)}$	-1095.8	-1012.1	65.7
$Mg_3N_{2(s)}$	-460.7	-406.0	90.0
$Mg(NO_3)_{2(s)}$	-790.7	-589.5	164.0
$Mg(NO_3)_2 \cdot 2H_2O_{(s)}$	-1409.2	NA†	soluble
$Mg(NO_3)_2 \cdot 6H_2O_{(s)}$	-2613.3	-2080.7	452.0
$MgS_{(s)}$	-346.0	-341.8	50.3
$MgSO_{4(s)}$	-1284.9	-1170.7	91.6
$MgSO_4 \cdot 2H_2O_{(s)}$	-1896.2	-1376.5	NA†
$MgSO_4 \cdot 4H_2O_{(s)}$	-2496.6	-2138.9	NA†
$MgSO_4 \cdot 6H_2O_{(s)}$	-3086.9	-2632.2	348.1
$MgSO_4 \cdot 7H_2O_{(s)}$	-3388.7	-2871.9	372.0
$Mg_3(PO_4)_2 \cdot 2H_2O_{(s)}$	-4022.9	NA†	7.61×10^{-5}
$Mg_2Si_{(s)}$	-77.8	-75.0	75.0

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$MgSiO_{3(s)}$	-1549.0	-11462.1	67.7
$Mg_2SiO_{4(s)}$	-2174.0	-2055.2	95.1
	Mn		
$Mn_{(s)}$	0	0	32.0
$Mn^{2+}_{(g)}$	2519.0	NA†	173.6
$Mn^{2+}_{(aq)}$	-233.0	-228.0	-74.6
$MnCl_{2(s)}$	-481.3	-440.5	118.2
$MnCl_2 \cdot H_2O_{(s)}$	-789.9	-696.2	174.1
$MnCl_2 \cdot 2H_2O_{(s)}$	-1092.0	-942.2	218.8
$MnCl_2 \cdot 4H_2O_{(s)}$	-1687.4	-1423.8	303.3
$MnBr_{2(s)}$	-384.9	-365.7	138.0
$MnBr_2 \cdot H_2O_{(s)}$	-705.0	NA†	
$MnBr_2 \cdot 4H_2O_{(s)}$	-1590.3	-1292.4	291.6
$MnI_{2(aq)}$	-331.0	-250.6	152.7
$MnI_2 \cdot 2H_2O_{(s)}$	-842.7	NA†	NA†
$MnI_2 \cdot 4H_2O_{(s)}$	-1438.9	NA†	NA†
$MnO_{(s)}$	-385.2	-362.9	59.7
$MnO_4{}^{-1}{}_{(aq)}$	-542.7	-449.4	191.0
$Mn_3O_{4(s)}$	-1387.8	-1283.2	155.6
$Mn_2O_{3(s)}$	-959.0	-881.2	110.5
$MnO_{2(s),pyrolusite}$	-520.0	-465.2	53.1
$Mn(OH)_{2(s)}$	-695.4	-615.0	99.2
$MnCO_{3(s)}$	-894.1	-816.7	85.8
$Mn(NO_3)_{2(s)}$	-576.3	-503.3	168.6
$Mn(NO_3)_2 \cdot 6H_2O_{(s)}$	-2371.9	-1809.6	NA†
$MnS_{(s)}$	-214.2	-218.4	78.2
$MnSO_{4(s)}$	-1065.2	-957.4	112.1
$MnSO_4 \cdot H_2O_{(s)}$	-1376.5	-1214.6	NA†
$MnSO_4 \cdot 4H_2O_{(s)}$	-2258.1	-1908.3	NA†
$MnSO_4 \cdot 5H_2O_{(s)}$	-2553.1	-2140.0	NA†
	Hg		
$Hg_{(l)}$	0	0	76.1
$Hg_{(g)}$	61.32	-178.6	146.0
$Hg^{2+}_{(g)}$	2890.4	NA†	174.9
$Hg_2{}^{2+}{}_{(aq)}$	172.3	153.6	84.5
$Hg_2F_{2(s)}$	-485.0	-435.6	160.7
$Hg_2Cl_{2(s),calomel}$	-265.2	-210.8	192.5
$HgCl_{2(s)}$	-224.3	-178.7	146.0
$Hg_2Br_{2(s)}$	-206.9	-181.1	218.0
$HgBr_{2(s)}$	-170.7	-153.1	172.0
$Hg_2I_{2(s)}$	-121.3	-111.0	233.5
$HgI_{2(s)}$	red	-105.4	-101.7
$HgO_{(s),red}$	-90.8	-58.6	70.26
$Hg(OH)_{2(aq)}$	-355.2	-274.9	142.3
$Hg_2(NO_3)_2 \cdot 2H_2O_{(s)}$	-868.2	-563.2	NA†
$HgS_{(s),black}$	-53.6	-47.7	88.3
$HgS_{(s),red}$	-58.2	-50.6	82.4
$Hg_2SO_{4(s)}$	-743.1	-625.9	200.7
$HgSO_{4(s)}$	-707.5	-590.0	145.0
	N		

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$NH_4OH_{(l)}$	-361.2	-254.1	165.6
$NH_4NO_{3(s)}$	-365.6	-184.0	151.1
$(NH_4)_2SO_{4(s)}$	-1180.9	-901.9	220.1
$NH_{3(g)}$	46.1	-16.5	192.3
	O		
$O_{2(g)}$	0	0	205
$O_{3(g)}$(ozone)	142.7	163.2	238.8
$OH^{-1}_{(aq)}$	-230.0	-157.2	-10.8
	P		
$P_{(s),white}$	0	0	41.1
$P_{4(g)}$	314.5	278.3	163.2
$PH_{3(g)}$	5.4	13.4	210.1
$PH_4I_{(s)}$	-69.9	0.8	123.0
$PF_{3(g)}$	-918.8	-897.5	273.1
$PF_{5(g)}$	-1595.8	NA†	281.0
$PCl_{3(l)}$	-319.7	-272.4	217.1
$PCl_{5(s)}$	-443.5	NA†	166.5
$POCl_{3(l)}$	-597.1	-520.9	222.5
$PBr_{3(l)}$	-184.5	-175.7	240.2
$PBr_{5(s)}$	-269.9	-NA†	decomp.
$POBr_{3(s)}$	-458.6	-430.5	NA†
$P_4O_{6(s)}$	-1640.1	NA†	decomp.
$P_4O_{10(s)}$	-2984.0	-2697.8	228.9
$P_2S_{5(s)}$	251.0	NA†	insoluble
	K		
$K_{(s)}$	0	0	64.2
$K^+_{(g)}$	514.3	481.2	154.4
$KF_{(s)}$	-567.3	-537.8	66.6
$KF \cdot 2H_2O_{(s)}$	-1163.6	-1021.6	155.2
$KCl_{(s)}$	-436.7	-409.2	82.6
$KClO_{3(s)}$	-397.7	-296.3	143.1
$KClO_{4(s)}$	-432.8	-303.2	151.0
$KBr_{(s)}$	-393.8	-380.7	95.9
$KBrO_{3(s)}$	-360.2	-271.2	149.2
$KBrO_{4(s)}$	-287.9	-174.5	170.1
$KI_{(s)}$	-327.9	-324.9	106.3
$KIO_{3(s)}$	-501.4	-418.4	151.5
$KIO_{4(s)}$	-467.2	-361.4	176.0
$K_2O_{(s)}$	-361.4	NA†	NA†
$KO_{2(s)}$	-284.9	-239.5	116.7
$KOH_{(s)}$	-424.8	-379.1	78.9
$KOH \cdot 2H_2O_{(s)}$	-1051.0	-887.4	151.0
$K_2CO_{3(s)}$	-1151.0	-1063.6	155.5
$KHCO_{3(s)}$	-963.2	-863.6	115.5
$KNO_{2(s)}$	-369.8	-306.6	152.1
$KNO_{3(s)}$	-494.6	-349.9	133.1
$KCN_{(s)}$	-113.0	-101.9	128.5
$KSCN_{(s)}$	-200.2	-178.3	124.3
$K_2S_{(s)}$	-380.7	-364.0	104.6
$K_2SO_{4(s)}$	-1437.8	-1321.4	175.6

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$KHSO_{4(s)}$	-1160.6	-1031.4	138.1
$KH_2PO_{4(s)}$	-1568.3	-1415.9	134.9
$KMnO_{4(s)}$	-837.2	-737.6	171.7
$K_2CrO_{4(s)}$	-1403.7	-1295.8	200.1
$K_2Cr_2O_{7(s)}$	-2061.4	-1882.0	291.2
$KAl(SO_4)_{2(s)}$	-2470.2	-2240.1	204.6
$KAl(SO_4)_2 \cdot 12H_2O_{(s)}$	-6061.8	-5141.7	687.4
$KCr(SO_4)_2 \cdot 12H_2O_{(s)}$	-5777.3	NA†	0.0441
$K_3Fe(CN)_{6(s)}$	-249.8	-129.7	426.1
$K_4Fe(CN)_{6(s)}$	-594.1	-453.1	418.8
$K_4Fe(CN)_6 \cdot 3H_2O_{(s)}$	-1466.5	-1169.0	593.7
	Rb		
$Rb^+_{(g)}$	490.1	NA†	164.2
$RbH_{(s)}$	-52.1	-32.2	NA†
$RbF_{(s)}$	-557.7	-523.4	82.1
$RbCl_{(s)}$	-435.3	-407.8	95.9
$RbClO_{3(s)}$	-402.9	-300.4	151.9
$RbClO_{4(s)}$	-437.2	-307.7	164.0
$RbBr_{(s)}$	-394.6	-381.8	110.0
$RbBrO_{3(s)}$	-367.3	-278.1	161.1
$RbI_{(s)}$	-333.8	-328.9	118.4
$RbIO_{3(s)}$	NA†	-426.3	NA†
$RbOH_{(s)}$	-418.2	NA†	84.1
$RbOH \cdot H_2O_{(s)}$	-748.9	NA†	
$RbOH \cdot 2H_2O_{(s)}$	-1053.2	NA†	
$Rb_2CO_{3(s)}$	-1136.0	-1051.0	181.4
$RbHCO_{3(s)}$	-936.2	-893.6	121.3
$RbNO_{3(s)}$	-495.1	-395.8	147.3
$Rb_2S_{(s)}$	-360.7	-339.0	134.0
$Rb_2SO_{4(s)}$	-1435.6	-1317.0	197.4
$RbHSO_{4(s)}$	-1158.9	-1030.1	NA†
	Sc		
$Sc^{3+}_{(g)}$	4627.0	NA†	156.3
$ScF_{3(s)}$	-1629.2	-1555.6	92.0
$ScCl_{3(s)}$	-925.1	-858.0	127.2
$Sc_2O_{3(s)}$	-1908.8	-1819.4	77.0
	Si		
$Si_{(s)}$	0	0	19.0
$SiH_{4(g)}$	34.3	56.9	204.5
$SiF_{4(g)}$	-1614.9	-1572.7	282.4
$SiCl_{4(l)}$	-687.0	-619.9	239.7
$SiCl_{4(g)}$	-657.0	-617.0	330.6
$SiBr_{4(l)}$	-457.3	-443.9	277.8
$SiBr_{4(g)}$	-415.4	-431.8	377.8
$SiO_{(g)}$	-99.6	-126.3	211.5
$SiO_{2(s),quartz}$	-910.9	-856.7	41.8
$SiO_{2(s)}$	-909.5	-855.9	42.7
$SiO_{2(s)}$	-909.1	-855.3	43.5
$SiC_{(s)}$	-62.8	-60.2	16.5
$SiS_{2(s)}$	-207.1	-175.3	66.9

Table 7.1 (continued) Standard thermodynamic functions at 1atm and 298K.

Substance	ΔH_f° (KJ/mol)	ΔG_f° (KJ/mol)	ΔS° (J/mol· K)
$Si^{4+}_{(g)}$	10428.5	NA†	229.8
Ag			
$Ag_{(s)}$	0	0	42.6
$Ag^{+}_{(g)}$	1019.2	NA†	167.2
$Ag^{+}_{(aq)}$	105.2	77.1	72.7
$AgF_{(s)}$	-204.6	-186.6	80.1
$AgF \cdot 2\,H_2O_{(s)}$	-800.8	-671.1	174.9
$AgF \cdot 4\,H_2O_{(s)}$	-1388.3	-1147.3	268.0
$AgCl_{(s)}$	-127.1	-109.8	96.2
$AgClO_{3(s)}$	-25.5	61.7	149.4
$AgClO_{4(s)}$	-31.1	77.0	NA†
$AgBr_{(s)}$	-100.4	-96.9	107.1
$AgBrO_{3(s)}$	-27.2	54.4	152.7
$AgI_{(s)}$	-61.8	-66.2	115.5
$Ag_2O_{(s)}$	-31.0	-11.2	121.3
$Ag_2CO_{3(s)}$	-505.8	-436.8	167.4
$AgNO_{3(s)}$	-124.4	-33.5	140.9
$AgCN_{(s)}$	146.0	156.9	107.2
$Ag_2S_{(s)}$	-29.4	-39.5	150.6
$Ag_2SO_{4(s)}$	-715.9	-618.5	200.4
$Ag_2CrO_{4(s)}$	-712.1	-621.7	216.7
Na			
$Na_{(s)}$	0	0	51.0
$Na^{+}_{(g)}$	609.0	NA†	147.9
$Na^{+}_{(aq)}$	-240.1	-261.9	59.0
$NaH_{(s)}$	-56.1	-33.5	40.0
$NaF_{(s)}$	-573.6	-543.5	51.5
$NaCl_{(s)}$	-411.2	-384.2	72.1
$NaClO_{3(s)}$	-365.8	-262.2	123.4
$NaClO_{4(s)}$	-383.3	-254.9	142.3
$NaBr_{(s)}$	-361.1	-349.0	86.8
$NaBr \cdot H_2O_{(s)}$	-951.9	-828.4	179.1
$NaBrO_{3(s)}$	-344.1	-242.8	128.9
$NaI_{(s)}$	-287.8	-286.1	98.5
$NaIO_{3(s)}$	-481.8	NA†	135.1
$NaIO_3 \cdot H_2O_{(s)}$	-779.5	-634.1	162.3
$NaIO_3 \cdot 5\,H_2O_{(s)}$	-1952.3	NA†	
$Na_2O_{(s)}$	-414.5	-375.5	75.1
$Na_2O_{2(s)}$	-510.9	-447.7	95.0
$NaOH_{(s)}$	-425.6	-379.5	64.5
$NaOH \cdot H_2O_{(s)}$	-734.5	-629.4	99.5
$Na_2CO_{3(s)}$	-1130.7	-1044.5	135.0
$Na_2CO_3 \cdot 10\,H_2O_{(s)}$	-4081.3	-3428.2	564.0
$NaHCO_{3(s)}$	-950.8	-851.0	101.7
$NaNO_{2(s)}$	-358.7	-284.6	103.8
$NaCN_{(s)}$	-87.5	-76.4	115.6
$Na_2S_{(s)}$	-364.8	-349.8	83.7
$Na_2SO_{4(s)}$	-1387.1	-1270.2	149.6
$Na_2SO_4 \cdot 10\,H_2O_{(s)}$	-4327.3	-3647.4	592.0
$NaHSO_{4(s)}$	-1125.5	-992.9	113.0
$Na_2S_2O_{3(s)}$	-1123.0	-1028.0	155.0
$Na_2S_2O_3 \cdot 5\,H_2O_{(s)}$	-2607.9	-2230.1	372.4
$Na_3PO_{4(s)}$	-1917.4	-1788.9	173.8
$Na_2SiO_{3(s)}$	-1554.9	-1461.0	113.8
$Na_2B_4O_{7(s)}$	-3291.1	-3096.2	189.2
$Na_2B_4O_7 \cdot 10\,H_2O_{(s)}$	-6288.6	-5516.6	585.5
$NaNH_{2(s)}$	-123.8	-64.0	76.9
Sr			
$Sr^{2+}_{(g)}$	1790.6	NA†	164.6
$SrF_{2(s)}$	-1216.3	-1164.8	82.1
$SrCl_{2(s)}$	-828.9	-781.2	114.9
$SrCl_2 \cdot H_2O_{(s)}$	-1136.8	-1036.4	172.0
$SrCl_2 \cdot 2\,H_2O_{(s)}$	-1438.0	-1282.0	218.0
$SrCl_2 \cdot 6\,H_2O_{(s)}$	-2623.8	-2241.2	390.8
$Sr(ClO_4)_{2(s)}$	-762.8	NA†	247.1
$SrBr_{2(s)}$	-717.6	-697.1	135.1
$SrI_{2(s)}$	-558.1	-562.3	159.0
$SrI_2 \cdot H_2O_{(s)}$	-886.0	NA†	
$SrI_2 \cdot 2\,H_2O_{(s)}$	-1182.4	NA†	
$SrI_2 \cdot 6\,H_2O_{(s)}$	-2388.6	NA†	
$Sr(IO_3)_{2(s)}$	-1019.2	-855.2	234.0
$SrO_{(s)}$	-592.0	-561.9	54.4
$Sr(OH)_{2(s)}$	-959.0	-869.4	88.0
$Sr(OH)_2 \cdot 8\,H_2O_{(s)}$	-3352.2	NA†	0.00655
$SrCO_{3(s)}$	-1220.1	-1104.4	97.1
$Sr(HCO_3)_{2(aq)}$	-1927.9	-1731.3	150.6
$Sr(NO_3)_{2(s)}$	-978.2	-780.1	194.6
$Sr(NO_3)_2 \cdot 4\,H_2O_{(s)}$	-2154.8	-1730.7	369.0
$SrS_{(s)}$	-453.1	-448.5	68.2
$SrSO_{4(s)}$	-1453.1	-1341.0	117.0
S			
$S_{(s),rhombic}$	0	0	31.8
$S^{2-}_{(aq)}$	33.1	85.8	-14.6
$SF_{4(g)}$	-774.9	-731.4	291.9
$SF_{6(g)}$	-1209.0	-1105.4	291.7
$SCl_{2(g)}$	-19.7	NA†	282.2
$SCl_{4(l)}$	-56.1	NA†	decomp.
$S_2Cl_{2(s)}$	-59.4	4.2	NA†
$SOCl_{2(l)}$	-245.6	-197.9	307.9
$SO_2Cl_{2(l)}$	-394.1	-305.0	216.7
$SO_{2(g)}$	-296.8	-300.2	248.1
$SO_{3(l)}$	-441.0	-368.4	95.6
$SO_{3(g)}$	-396.0	-370.0	256.0
$S_{(g)}$	278.8	238.3	167.8
$S_{2(g)}$	128.4	79.3	228.1
$S_{8(g)}$	102.3	49.7	430.9
Sn			
$Sn_{(s,white)}$	0	0	51.6
$Sn^{2+}_{(g)}$	2434.9	NA†	168.4
$Sn^{2+}_{(aq)}$	-8.8	-27.2	-17.0
$Sn^{4+}_{(g)}$	9323.2	NA†	168.4
$SnH_{4(g)}$	162.8	188.2	227.6
$SnCl_{2(s)}$	-325.1	NA†	1.42
$SnCl_2 \cdot 2\,H_2O_{(s)}$	-921.3	-787.8	NA†
$SnCl_{4(l)}$	-511.3	-440.2	258.6
$SnBr_{2(s)}$	-243.5	-250.6	146.0
$SnBr_{4(s)}$	-377.4	-350.2	264.4
$SnBr_4 \cdot 8\,H_2O_{(s)}$	-276.8	NA†	
$SnI_{2(s)}$	-143.5	-145.2	168.6
$SnO_{(s)}$	-285.8	-256.9	56.5
$SnO_{2(s)}$	-580.7	-519.7	52.3
$SnS_{(s)}$	-100.0	-98.3	77.0
$Sn(SO_4)_{2(s)}$	-1629.2	-1443.0	155.2
Ti			
$Ti^{2+}_{(g)}$	2450.6	NA†	
$Ti^{3+}_{(g)}$	9290.2	NA†	
$TiH_{2(s)}$	-119.7	-80.3	29.1
$TiCl_{2(s)}$	-513.8	-464.4	87.4
$TiCl_{3(s)}$	-720.9	-653.5	139.7
$TiCl_{4(s)}$	-804.2	-737.2	252.3
$TiBr_{2(s)}$	-402.0	-375.0	130.1
$TiBr_{3(s)}$	-548.5	-523.8	176.6
$TiBr_{4(s)}$	-616.7	-589.5	243.5
$TiI_{2(s)}$	-263.0	-270.1	147.7
$TiI_{4(s)}$	-375.7	-371.5	249.4
$TiO_{2(s)}$	-939.7	-884.5	49.9
$Ti_2O_{3(s)}$	-1520.9	-1434.3	78.9
W			
$W^{+}_{(g)}$	1625.9	NA†	
$WF_{6(l)}$	-1747.7	-1631.4	251.5
$WCl_{2(s)}$	-255.0	-213.6	130.2
$WCl_{4(s)}$	-467.0	-303.1	344.5
$WCl_{6(s)}$	-682.5	-548.9	254.0
$WBr_{6(s)}$	-348.5	-328.0	472.0
$WO_{3(s),wolfamite}$	-842.9	-764.1	75.9
$WS_{2(s)}$	-209.0	NA†	84.0
$WC_{(s)}$	-40.5	-40.2	35.6
U			
$UF_{6(g)}$	-2112.9	-2029.3	379.7
$UCl_{2(s)}$	-75.3	-80.3	79.0
$UCl_2O_{2(s)}$	-1263.1	-1159.0	150.5
$UO_{2(s)}$	-1129.7	-1075.3	77.8
$UO_{3(s)}$	-1263.6	-1184.1	98.6
$U_2C_{3(s)}$	-205.0	-201.0	105.0
$UO_2(NO_3)_{2(s)}$	-1377.4	-1142.7	276.1
$UO_2(NO_3)_2 \cdot 6\,H_2O_{(s)}$	-3197.8	-2615.0	505.6
$US_{2(s)}$	-502.0	-531.7	110.5
V			
$V^{2+}_{(g)}$	2590.5	NA†	169.4
$V^{3+}_{(g)}$	5430.5	NA†	171.5
$V^{4+}_{(g)}$	9943.3	NA†	169.3
$VF_{4(s)}$	-1403.3	NA†	
$VF_{5(l)}$	-1480.3	-1373.2	175.7
$VF_{5(g)}$	-1433.8	-1369.8	320.8
$VCl_{2(s)}$	-452.0	-406.0	97.1
$VCl_{3(s)}$	-580.7	-511.3	131.0
$VCl_{4(l)}$	-569.4	-503.7	255.2
$VBr_{2(s)}$	-365.3	NA†	126.0
$VBr_{3(s)}$	-433.5	NA†	142.0
$VBr_{4(g)}$	-336.8	NA†	335.0
$VI_{2(s)}$	-251.5	NA†	143.1
$VO_{(s)}$	-431.8	-404.2	38.9
$V_2O_{3(s)}$	-1228.0	-1139.3	98.3
$V_2O_{5(s)}$	-1550.6	-1419.6	131.0
Xe			
$XeF_{2(s)}$	-133.9	-62.8	133.9
$XeF_{4(s)}$	-261.5	-121.3	146.4
$XeF_{6(s)}$	-380.7	NA†	
$XeO_{3(s)}$	401.7	NA†	
Zn			
$Zn_{(s)}$	0	0	41.6
$Zn^{2+}_{(g)}$	2782.7	NA†	160.9
$Zn^{2+}_{(aq)}$	-153.9	-147.1	-112.1
$ZnF_{2(s)}$	-764.4	-449.5	73.7
$ZnCl_{2(s)}$	-415.1	-369.4	111.5
$ZnBr_{2(s)}$	-328.7	-312.1	138.5
$ZnI_{2(s)}$	-208.0	-208.9	161.1
$ZnO_{(s)}$	-348.3	-318.3	43.6
$ZnCO_{3(s)}$	-812.8	-731.6	82.4
$Zn(NO_3)_{2(s)}$	-483.7	NA†	
$Zn(NO_3)_2 \cdot 6\,H_2O_{(s)}$	-2306.6	-1773.1	456.9
$ZnS_{(s),wurtzite}$	-192.6	-187.0	57.7
$ZnS_{(s),blende}$	-206.0	-201.3	65.3
$ZnSO_{4(s)}$	-982.8	-874.5	119.7
$ZnSO_4 \cdot 7\,H_2O_{(s)}$	-3077.8	-2563.1	388.7